

Impact of Information Security Awareness Training on Knowledge, Attitude, and Behavior: A K-12 Case Study

Michael S. Robbins

Christopher Robbins

Follow this and additional works at: <https://digitalcommons.kennesaw.edu/jcerp>



Part of the [Information Security Commons](#), [Management Information Systems Commons](#), and the [Technology and Innovation Commons](#)

Impact of Information Security Awareness Training on Knowledge, Attitude, and Behavior: A K-12 Case Study

Abstract

Abstract— Information security breaches remain a serious threat across all sectors, often exploiting human factors rather than technical flaws. This study examines how a structured Information Security Awareness (ISA) training program influences employees' knowledge of security policies, attitudes towards those policies, and self-reported security behaviors within a K-12 educational environment. A quantitative pre-test/post-test design was employed with 201 staff members (administrators, teachers, and support personnel) in a public school district. Participants completed the Human Aspects of Information Security Questionnaire (HAIS-Q) before and after undergoing an interactive cybersecurity training program. Statistical analysis revealed a significant improvement in information security knowledge, attitudes, and behavior following the training ($p < 0.001$ for all domains). Additionally, strong positive correlations were observed among knowledge, attitude, and behavior both before and after training. These results indicate that well-designed ISA training can effectively enhance the human aspects of cybersecurity, reinforcing employees' understanding of policies and encouraging safer security practices. While conducted in a single K-12 district, the findings suggest the potential applicability of the KAB model and interactive training approaches to other educational contexts and organizational sectors, though results may vary depending on organizational culture and demographic factors. The study also highlights that the improvements were measured in the short term; future research should evaluate long-term retention and behavioral maintenance. Implications for broader organizational practices and recommendations for addressing demographic and contextual differences in future studies are discussed.

Keywords

Information Security Awareness, K-12 Education, Cybersecurity Training, Knowledge Attitude Behavior Model, HAIS-Q, Human Factors

Impact of Information Security Awareness Training on Knowledge, Attitude, and Behavior: A K-12 Case Study

Michael S Robbins
Innovation and Technology Services
Westminster School District
Westminster, CA, USA
mrobbins@wsdk8.us
ORCID: 0009-0001-6913-7246

Christopher B Robbins
College of Pharmacy
South Dakota State University
Brookings, SD, USA
Christopher.Robbins@sdstate.edu
ORCID: 0009-0000-1117-8814

Abstract— *Information security breaches remain a serious threat across all sectors, often exploiting human factors rather than technical flaws. This study examines how a structured Information Security Awareness (ISA) training program influences employees' knowledge of security policies, attitudes towards those policies, and self-reported security behaviors within a K-12 educational environment. A quantitative pre-test/post-test design was employed with 201 staff members (administrators, teachers, and support personnel) in a public school district. Participants completed the Human Aspects of Information Security Questionnaire (HAIS-Q) before and after undergoing an interactive cybersecurity training program. Statistical analysis revealed a significant improvement in information security knowledge, attitudes, and behavior following the training ($p < 0.001$ for all domains). Additionally, strong positive correlations were observed among knowledge, attitude, and behavior both before and after training. These results indicate that well-designed ISA training can effectively enhance the human aspects of cybersecurity, reinforcing employees' understanding of policies and encouraging safer security practices. While conducted in a single K-12 district, the findings suggest the potential applicability of the KAB model and interactive training approaches to other educational contexts and organizational sectors, though results may vary depending on organizational culture and demographic factors. The study also highlights that the improvements were measured in the short term; future research should evaluate long-term retention and behavioral maintenance. Implications for broader organizational practices and recommendations for addressing demographic and contextual differences in future studies are discussed.*

Keywords— *Information Security Awareness, K-12 Education, Cybersecurity Training, Knowledge Attitude Behavior Model, HAIS-Q, Human Factors*

I. INTRODUCTION

Modern organizations face relentless cyber threats that can result in severe data breaches and financial losses. Whether a breach affects millions (as in the 2014 Target Corp. incident) or a handful of individuals, organizations must take proactive measures to prevent such incidents. Not only is this critical from a privacy standpoint, but also from a financial one – the average data breach can impose an economic cost of around \$2.5 million. Beyond external attackers, insiders and end-users have emerged as one of the greatest vulnerabilities in an

organization's cybersecurity defenses. In today's landscape, the "human element" – employees' mistakes or non-compliance – often allows attackers to circumvent technical safeguards. Lack of awareness and understanding of information security best practices among staff can therefore dramatically increase an organization's risk exposure.

To mitigate this internal risk, many organizations have turned to information security awareness (ISA) training programs. Effective ISA training seeks to educate employees about security policies, expected behaviors, and the importance of following procedures, thereby addressing the cognitive and behavioral gaps that technology alone cannot fix. Prior research has found that improving employees' knowledge of security policies and procedures can lead to more positive attitudes toward those policies, which in turn encourage safer behaviors (such as compliance and incident reporting). This relationship is often described by the Knowledge, Attitude, Behavior (KAB) model, which posits that increasing one's knowledge can favorably influence one's attitude, ultimately shaping one's behavior. The KAB model underpins the Human Aspects of Information Security Questionnaire (HAIS-Q) framework developed by Parsons et al. (2014) to measure security awareness in organizations. According to this model, training and education are most effective when they not only convey *what* is expected (knowledge of policies) but also *why* it is important (instilling a positive attitude), thereby motivating users to *act* securely (behavior).

While numerous studies have examined security awareness training in corporate and government settings, the K-12 education sector has received comparatively less attention. Public school districts manage extensive personal data on students, families, and employees, making them attractive targets for cyberattacks. Indeed, K-12 institutions are facing an increasing volume of phishing and malware attacks in recent years. However, resource constraints and competing priorities often mean that cybersecurity training in schools is

minimal or inconsistent. There is a clear need for research on how ISA training impacts school staff, and whether lessons learned in other industries apply to educational environments. If a security awareness program can measurably improve the knowledge, attitudes, and behaviors of K-12 employees, it could greatly enhance the protection of sensitive student and staff data.

Purpose of this study: To address this gap, we conducted a case study of an information security awareness training intervention in a California K-12 school district. We evaluated the training's impact on three key outcomes – employees' knowledge of security policies, their attitude toward following these policies, and their self-reported behavior in handling security threats – using the HAIS-Q instrument. Specifically, we asked: (1) Does an increase in security policy knowledge correspond to a more positive attitude toward security compliance? (2) Does a more positive security attitude, in turn, relate to better security behavior (e.g., willingness to report potential incidents)? and (3) Does greater knowledge directly contribute to improved security behavior? These questions align with the KAB model's chain of influence (knowledge → attitude → behavior) and reflect the hypotheses that greater knowledge of and positive attitude toward security policies will correlate with safer, more proactive security behaviors.

In the remainder of this paper, we present related work on security awareness and the KAB model, describe our methodology and data collection, report the results of the pre- and post-training assessments, and discuss the implications of the findings for K-12 institutions and other sectors. Through this work, we aim to demonstrate how a comprehensive ISA training program can strengthen the human element of cybersecurity in education and beyond.

II. REVIEW OF LITERATURE

Human Factors in Information Security: It is widely acknowledged that technology alone cannot ensure information security; the behaviors of computer users are a critical factor. Studies have shown that insider threats and human errors contribute significantly to security incidents, motivating the development of metrics and tools to quantify human-centric security vulnerabilities. One such effort is the creation of structured questionnaires to assess information security awareness (ISA). For example, Kruger and Kearney (2006) defined ISA as “the degree to which an organization's members understand the importance of information security and their individual responsibilities. They proposed measuring ISA through components like knowledge of security issues, attitudes toward security, and self-reported behavioral practices. These components form the basis of many later frameworks.

The KAB Model and HAIS-Q: The Knowledge–Attitude–Behavior (KAB) model originates from social psychology and has been applied in various domains (e.g., public health campaigns) to explain how educating individuals can lead to attitude and behavior change. In the information security context, Parsons et al. adopted the KAB model to develop the Human Aspects of Information Security Questionnaire (HAIS-Q). The HAIS-Q, first introduced by Parsons et al. (2014), is a survey instrument designed to measure employees' knowledge of security policies, their attitudes toward those policies, and their self-reported security behaviors. In their initial study with 500 Australian office workers, Parsons et al. found that an employee's knowledge of information security policies and procedures had a direct positive influence on their attitude towards those policies, and together knowledge and attitude explained a large portion of the variance in self-reported behavior. In other words, employees who were more aware of security expectations tended to view security rules more favorably, and those with positive attitudes were more likely to behave securely. This provided empirical support for the KAB model in the cybersecurity domain.

Subsequent studies further validated the HAIS-Q and the KAB model. Parsons et al. (2017) expanded their research to multiple industry contexts and confirmed a strong positive correlation among knowledge, attitude, and behavior measures, demonstrating the instrument's reliability and validity. Other researchers have applied the HAIS-Q or similar KAB-based assessments in different environments – for instance, Hadlington & Parsons (2017) examined factors like “cyberloafing” (personal internet use at work) and found it impacted HAIS-Q scores. Beyond cybersecurity, the KAB approach has been used to study topics such as medical professionals' awareness of elder abuse, where it helped identify knowledge gaps and guided effective training programs. These examples underscore the versatility of the KAB model in shaping training interventions: the key is ensuring the training not only imparts factual knowledge but also addresses attitudes and norms, thereby motivating behavior change.

Information Security Awareness Training: Prior work emphasizes that how training is delivered can be as important as the content itself. Traditional approaches like annual slide presentations or lectures may raise awareness in the short term but often have limited long-term impact on behavior. Interactive and engaging training formats tend to be more effective. For instance, Parsons et al. (2014) suggested that training should outline not just the “what” (policies and rules) but also the “why” behind security practices, to positively influence attitudes. Many organizations now use computer-based training modules that include videos, quizzes, and simulated phishing emails to actively involve employees in learning. Such dynamic programs can keep users engaged and

reinforce retention better than passive instruction. There is evidence that when employees are engaged and understand the relevance of security measures, they are more likely to internalize the practices and comply with security policies on the job.

In the education sector, studies specifically focusing on K-12 or higher education staff are relatively scarce compared to corporate settings. However, the need for robust ISA training in schools is growing. Schools are increasingly targeted by phishing scams, ransomware attacks, and other threats. A successful cyberattack on a school system can disrupt operations and compromise large volumes of personal information (students' names, academic records, health data, etc.). K-12 institutions also face challenges such as limited IT staffing and budget, and a diverse user base (educators, administrative staff, IT staff) with varying levels of technical expertise. Despite these challenges, schools have a communal incentive to improve security awareness, as they often share information about emerging threats and best practices with each other in the interest of protecting students. Our study contributes to the literature by examining a K-12 case in depth, thereby extending the application of the HAIS-Q based KAB framework into the education domain. We aimed to see if findings from prior KAB research in corporate contexts – such as the positive impacts of interactive training and the interrelationships among knowledge, attitude, and behavior – hold true among school district employees. Confirmation of this would suggest that similar training strategies could be employed in other sectors that have comparable needs for improving the human aspect of cybersecurity.

III. METHODOLOGY

Study Design and Participants

We conducted a quantitative field study at Saddleback Valley Unified School District (SVUSD) in Southern California – a K-12 public school district with nearly 28,000 students and over 2,000 employees. The study followed a pre-test/post-test single-group design. All district employees were invited to participate, and those who volunteered completed a survey before and after an information security training program. A total of $N = 201$ staff members participated in the study, representing a broad cross-section of job roles. This included district and site administrators, classified managers and support staff, certificated teachers, and other school employees. Participation was voluntary and responses were anonymized. Given the organizational context (a public school system), we obtained necessary approvals from district leadership and site principals to conduct the survey during the study period. All participants provided informed consent, and the study complied with relevant research ethics standards.

Procedure and Training Intervention

The study was conducted in mid-2019. Baseline (pre-training) data collection took place in June–July 2019. Participants received an email with a link to an online HAIS-Q survey (administered via the Qualtrics platform) and were asked to complete the survey to assess their current knowledge, attitudes, and behaviors regarding information security. Immediately following the baseline survey, the district rolled out a comprehensive Information Security Awareness training program to all staff. The training content was delivered through a commercial e-learning platform (Sophos Phish Threat) that provided interactive modules focused on various cybersecurity topics (e.g., phishing email recognition, safe internet use, password security). The program consisted of several short interactive videos and quizzes that users had to complete. This dynamic format was chosen over a static slideshow or lecture in order to maximize engagement and knowledge retention, as recommended by prior research. The training emphasized not only the correct security practices (what to do or not do) but also explained the rationale behind them (why it matters), in an effort to influence participants' attitudes positively towards compliance.

Upon completing the training modules – which most participants did within a few weeks – employees were then asked to complete the post-training HAIS-Q survey. The post-test survey was administered from August to September 2019, roughly 4–6 weeks after the training, to allow participants time to internalize the material. The same HAIS-Q instrument was used, with additional questions included to capture demographic information (such as the participant's department/work location and job classification) and to note whether the participant chose not to disclose their department. By matching pre- and post-survey responses, we evaluated changes in each individual's scores from before to after the training program.

Measurement Instrument: HAIS-Q

The Human Aspects of Information Security Questionnaire (HAIS-Q) was the primary measurement tool. The HAIS-Q is a validated survey instrument that assesses three constructs – Knowledge, Attitude, and Behavior – in relation to information security, following the KAB model. In its full form, the HAIS-Q covers multiple domains of security behavior. The version used in this study was a 63-item questionnaire targeting seven key focus areas of information security awareness:

- Internet use (e.g., recognizing safe vs. risky web practices)
- Email use (e.g., identifying phishing emails and handling email attachments)

- Social media use (e.g., understanding privacy settings and what not to share)
- Password management (including creating strong passwords and locking workstations)
- Incident reporting (e.g., knowing how and when to report a security incident)
- Information handling (proper handling of sensitive data, both digital and physical)
- Mobile device security (e.g., securing laptops, smartphones, USB drives)

Within each focus area, the survey included questions evaluating the participant's *knowledge* of correct practices, their *attitude* (or beliefs) about following those practices, and their self-reported *behavior* (frequency or consistency of doing so). Responses were given on a Likert-type scale (e.g., 1 = strongly disagree to 5 = strongly agree for knowledge/attitude items, and analogous frequency scales for behavior items). Each construct (Knowledge, Attitude, Behavior) thus had a composite score aggregated over all relevant questions. For example, a participant's "Knowledge" score represented their overall knowledge across all seven security domains. Higher scores indicated better security awareness in that construct.

The HAIS-Q has established content validity and internal consistency in prior studies. In our implementation, we conducted reliability analysis on the survey responses and found that Cronbach's alpha values for the Knowledge, Attitude, and Behavior scales all exceeded 0.7, indicating good internal consistency in our sample. This suggests that the survey items for each construct were measuring a cohesive underlying factor. We used each participant's summed (or average) score per construct for statistical analysis.

Data Analysis

We analyzed the survey data using IBM SPSS Statistics (v22). Our analysis addressed two aspects: (1) pre-training vs. post-training differences in the Knowledge, Attitude, and Behavior scores, to evaluate the impact of the training intervention; and (2) interrelationships among Knowledge, Attitude, and Behavior, both before and after training, as well as any differences in outcomes across demographic groups.

Given that the data were primarily ordinal Likert-scale scores and not strictly normally distributed, we used non-parametric statistical tests. To test for significant changes in scores from pre-test to post-test, we employed the Wilcoxon signed-rank test (the non-parametric equivalent of a paired t-test). This test is appropriate for comparing two related samples (in this case, each participant's pre-training score vs. their post-training

score). We performed separate Wilcoxon tests for the Knowledge, Attitude, and Behavior constructs. The null hypothesis for each was that there is no difference between pre- and post-training scores, and the alternative hypothesis was that the post-training scores are higher.

To assess the relationships between Knowledge, Attitude, and Behavior scores, we calculated Spearman's rho correlation coefficients. We first examined the baseline (pre-training) correlations between each pair of constructs (Knowledge–Attitude, Knowledge–Behavior, and Attitude–Behavior). We then did the same for the post-training scores. Spearman's rho (ρ) is a rank-based correlation measure suitable for ordinal data and was used here to determine how closely aligned these dimensions were. We expected positive correlations, consistent with the KAB model (e.g., employees who know more about security are likely to have a more favorable attitude toward it, and those with positive attitudes are more likely to report secure behaviors). Additionally, Spearman correlation was used to check if any demographic factors had an influence – for example, we looked at whether belonging to a particular department or job category correlated with different baseline scores or different degrees of improvement. We also conducted Kruskal-Wallis tests to compare groups (such as differences between departments) when applicable, though as we report below, few group differences were observed.

All hypothesis tests were conducted with a significance level of $\alpha = 0.05$ (with Bonferroni adjustments made if multiple comparisons were considered together). Effect sizes for Wilcoxon tests (when significant) were evaluated by converting the Z statistic to an r value (by $r = Z/\sqrt{N}$) to interpret the magnitude of the change. The results are presented with relevant test statistics and p-values. **Table 1** below summarizes the statistical test results for the main pre- vs. post-training comparisons.

IV. RESULTS AND DISCUSSION

Participant Profile

A total of 201 employees submitted complete responses to both the pre-training and post-training surveys. This sample included representatives from virtually all functional groups in the district. Approximately half of the respondents were teachers, with the remainder composed of school administrators, district office staff, classified support staff, and others (including counseling, library, and IT personnel). This distribution reflects the overall employee composition of the district, lending confidence that the findings are broadly relevant across the organization. Participants' lengths of service and technical backgrounds varied, but all had daily access to computers and email as part of their jobs. Prior to this study, the district had not conducted any formal

cybersecurity training for staff, so the baseline results offer a first-time glimpse into the staff's security awareness levels.

Pre-training Baseline Assessment

The initial HAIS-Q survey revealed some important baseline relationships. Even before the training, the three security awareness dimensions (knowledge, attitude, behavior) were found to be strongly positively correlated with each other. For example, employees' knowledge scores were highly correlated with their attitude scores (Spearman $\rho \approx 0.84$), and both were strongly correlated with their self-reported behavior scores ($\rho \approx 0.84$ – 0.89), with all correlations significant at the 0.01 level. This means that in our sample, staff who knew more about the district's security policies also tended to hold more positive attitudes toward those policies, and those with positive attitudes were in turn more likely to report engaging in recommended security behaviors. These findings are consistent with the KAB model expectations and with prior studies by Parsons et al., which found knowledge and attitude to be key predictors of self-reported security behavior.

Despite the generally high correlations, there was room for improvement in each area. Qualitative feedback and some survey items indicated that certain practices – for instance, reporting suspicious emails or regularly updating passwords – were not universally followed, suggesting gaps in behavior. Likewise, not all staff were fully aware of the latest security policies or the rationale behind them (knowledge and attitude gaps). These observations reinforced the need for the training intervention.

Post-training Improvements

After the ISA training program was implemented, participants' HAIS-Q scores showed marked improvements. Table 1 summarizes the comparison of pre- vs. post-training scores for each of the three constructs (Knowledge, Attitude, Behavior). We found a statistically significant increase in all three areas after the training (Wilcoxon signed-rank tests, $p < 0.001$ for each comparison). In practical terms, employees demonstrated greater knowledge of information security (e.g., higher awareness of policies and best practices), reported a more positive attitude toward complying with security procedures, and indicated more frequent or consistent secure behaviors following the training.

Table 1. Pre- vs. Post-Training Differences in Security Awareness (HAIS-Q Scores) (Wilcoxon signed-rank tests, $N = 201$).

Domain	Wilcoxon (Pre vs. Post)	p -value	Outcome
Knowledge	–7.309	< 0.001	Significant increase (post > pre)
Attitude	–3.783	< 0.001	Significant increase (post > pre)
Behavior	–5.198	< 0.001	Significant increase (post > pre)

As shown in Table 1, the largest change was observed in the Knowledge domain ($Z = -7.309$), indicating that the training had a very pronounced effect on employees' understanding of security policies and best practices. This is perhaps unsurprising, as imparting knowledge is the most direct goal of any training program. Many participants who initially had limited familiarity with certain topics (for example, how to recognize a phishing email or the importance of two-factor authentication) gained substantial knowledge as reflected in their post-training responses. The Behavior domain also saw a strong improvement ($Z = -5.198$), suggesting that employees were more likely to report engaging in secure behaviors after the training. For instance, more staff reported that they would not click on suspicious links, would report incidents promptly, and would follow guidelines for handling sensitive information. The improvement in Attitude, while statistically significant ($Z = -3.783$), was somewhat smaller in magnitude compared to knowledge and behavior. This could indicate that many staff already had reasonably positive attitudes toward security at baseline (e.g., they agreed in principle that following security rules is important), but there was still a measurable positive shift post-training. Even a modest uplift in attitude is valuable, as positive attitude can reinforce ongoing compliance and peer support for security initiatives.

It is important to note that the strong inter-correlations among knowledge, attitude, and behavior persisted after the training. In the post-training data, these constructs remained highly correlated (with Spearman ρ values in the ~ 0.87 – 0.91 range, all $p < 0.01$). This continued alignment suggests that the training uplifted all aspects of security awareness in tandem: those who learned the most (knowledge gain) also tended to improve their attitudes and behaviors the most. It reinforces the idea that knowledge gains can translate into attitudinal and behavioral changes, which is exactly what the KAB model predicts.

Furthermore, our analysis found no significant differences in training effectiveness across different employee groups. Improvements in scores were consistent regardless of the participant's department or job classification. For example, teachers, administrators, and support staff all showed comparable gains in their HAIS-Q scores. There was no evidence that one group benefited while another did not; all groups appeared to have been positively impacted by the program. This uniform outcome is encouraging, as it indicates the training content was universally applicable and accessible to staff in varied roles. It also aligns with prior findings that the HAIS-Q instrument and the underlying KAB model can be applied to diverse employee populations with similar effectiveness.

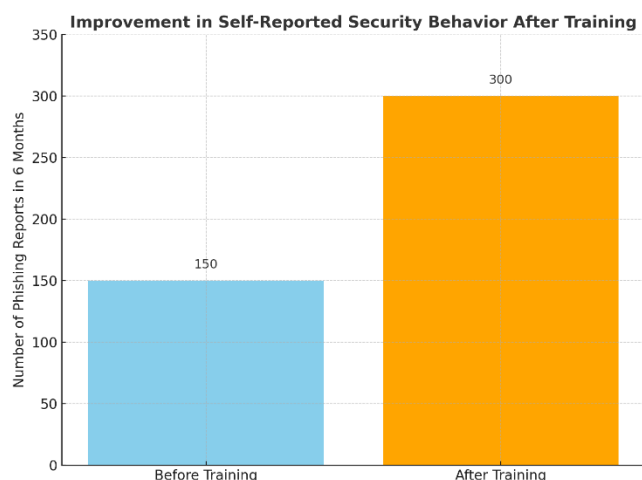


Figure 1: Improvement in self-reported security behavior after training. The chart shows the number of phishing email incidents reported by employees in a six-month period before the security awareness training (left, using the old helpdesk ticket method) compared to after the training when an integrated email reporting tool was introduced (right). After the training intervention, employees reported over 300 phishing-suspicious emails in six months – more than double the approximately 150 reports made in the comparable period before training. This real-world outcome illustrates a tangible increase in proactive security behavior, corroborating the survey results. Not only did employees say they would behave more securely (per the HAIS-Q responses), but many indeed took action by reporting threats at a much higher rate once they had been trained and given a convenient reporting mechanism. This significant jump in incident reporting enhances the organization's ability to respond to threats quickly and reflects a positive shift in the security culture.

Additional Observations

Beyond the core survey metrics, some qualitative feedback and follow-up observations provided context to the results. Many participants reported that the interactive nature of the

training modules (with scenario-based quizzes and immediate feedback) helped them better understand the consequences of unsafe behaviors, which in turn made them more confident in practicing safe computing. A number of staff members indicated that they had previously underestimated certain threats (e.g., thought that “phishing wouldn’t happen to our school”), but the training opened their eyes to the very real risks targeting education institutions. This change in perspective is an attitudinal shift that likely contributed to the increased willingness to report incidents (as evidenced in Figure 1). The introduction of the one-click email reporting tool (a plugin in Outlook) also lowered the barrier to reporting, allowing employees to translate their improved awareness into action quickly. The synergy of training and tool implementation appears to have been effective: employees were not only *taught* to be vigilant (training), but also *empowered* with an easy way to act on that vigilance (technical tool), resulting in a measurable behavior change.

No notable adverse effects were observed during the study. A small number of participants did not complete the post-training survey (they were excluded from analysis), but among those who did, the response patterns suggested honest effort and attention (e.g., we did not see evidence of straight-line answering or other invalid responses). One constraint faced was timing – because the training and surveys had to be completed within a summer vacation window, some staff felt hurried. This was noted as a limitation (discussed later), as more time might have led to even greater absorption of the material. Nonetheless, the majority completed the program and surveys in the allotted time.

In summary, the results provide strong evidence that the ISA training program achieved its intended outcomes: increasing the cybersecurity knowledge of K-12 staff, fostering more positive attitudes toward security compliance, and encouraging safer security behaviors (both in self-reports and in actual practice, as seen in reporting rates). These outcomes answer our research questions in the affirmative – *knowledge gains were associated with improved attitudes, and together these corresponded with improved security behaviors*. We next turn to interpreting what these findings mean for the school district and for other organizations aiming to bolster security through human-focused initiatives.

Discussion

The findings of this study demonstrate the efficacy of a well-designed information security awareness (ISA) training program in a K-12 educational setting. After the training intervention, employees showed statistically significant improvements in their security-related knowledge, attitudes, and behaviors. This aligns with prior research in corporate contexts and provides new evidence that those principles apply similarly in schools. Below, we discuss the implications

of these results in light of the original hypotheses and existing literature, consider the practical significance of the changes observed, and outline the lessons learned for broader application.

Validation of the KAB Model: Our results strongly support the Knowledge–Attitude–Behavior (KAB) model in the context of security awareness. Consistent with Parsons et al. (2014), we found that increasing employees’ knowledge of security policies and best practices correlates with more positive attitudes toward those policies, which in turn correlates with better self-reported security behavior. In our pre-training data, knowledge and attitude were already highly correlated ($\rho \approx 0.84$), suggesting that staff who understood the importance of security tended to agree with the need for security rules. More critically, after the training, these correlations remained high or even increased slightly (knowledge–behavior correlation rose to $\rho \approx 0.91$), indicating that those who learned the most from the training also adopted the most positive attitudes and were most likely to change their behavior. In practical terms, an employee who, for example, gained knowledge that “USB drives should be encrypted to protect data” was not only more aware of that fact, but also more likely to internalize the importance of the practice (attitude) and actually use encryption or avoid unencrypted USB use (behavior). This chain of influence, from knowledge gain to attitude shift to behavioral change, is exactly what the KAB model predicts and what our results confirm.

Our findings align with earlier studies that argued knowledge influences attitude more strongly than direct behavior. Indeed, Parsons et al. noted that training is most effective when it addresses why policies are important, thereby shaping attitudes that sustain behavior change. In our training program, we included the rationale and consequences (for instance, demonstrating how a phishing attack could compromise student data) rather than just do’s and don’ts. The positive shift in attitudes post-training suggests this approach worked – employees were not just rote learning rules, but actually buying into the importance of compliance. Several participants commented that seeing examples of real attack scenarios in the training videos made them realize that “it could happen here” and that following security procedures is part of their responsibility to the school community. This attitudinal buy-in is crucial because employees with a positive attitude toward security are more likely to adhere to security policies even when unsupervised.

Effectiveness of Interactive Training: The success of the training program highlights the value of interactive and engaging training methods. Our program featured interactive quizzes, videos, and phishing email simulations rather than passive lectures or slide decks. This aligns with best practices in security awareness training literature, which suggest that engaging formats improve retention and encourage behavior change. Employees remained interested and could

immediately apply what they learned, reinforcing their knowledge. The high knowledge gains measured reflect that participants absorbed the material well. Moreover, interactive training likely had a motivating effect; it kept users involved and signaled that the organization is actively investing in them.

The immediate availability of a reporting tool (the phishing email add-in) after the training was a strategic move. It provided an outlet for the newly heightened awareness – employees could act on their knowledge/attitude by reporting suspicious emails with one click. This convenience, combined with the knowledge of why reporting matters, led to the dramatic increase in reports illustrated in Figure 1. It suggests that training outcomes improve when paired with user-friendly security tools and processes. In practice, organizations should not only train users but also make it as easy as possible for them to act securely. Training and tools together create an environment where secure behavior is both understood and facilitated.

Implications for K-12 and Other Sectors: A key contribution of this study is demonstrating that K-12 school staff can benefit from the same structured ISA training historically used in corporate environments. Often, school districts assume that cybersecurity training is more relevant for industries like finance or healthcare. Our results counter that notion – the threats are very real for education, and school employees respond positively to training. There is nothing inherent in the K-12 context that diminishes the effectiveness of a good security awareness program. K-12 institutions may even have a collaborative advantage: schools often share knowledge and resources in a non-competitive way, allowing successful training initiatives to be adapted and rolled out in other districts.

Generalizability: While this study was conducted within a single California K-12 school district, the underlying principles of the KAB model and interactive ISA training are likely applicable to other educational settings and organizational sectors. The positive outcomes observed across diverse employee roles suggest that well-designed training programs can be effective even in heterogeneous populations. However, caution is warranted when extending these results to organizations with different cultural, technological, or regulatory environments. Factors such as district size, IT infrastructure, prior experience with cybersecurity, and leadership support could influence the effectiveness of similar interventions. Future studies could replicate this design in multiple districts or corporate environments to examine contextual factors and determine which elements of the training are universally effective versus context-specific.

Short-term Measurement Limitation: A notable limitation is that post-training outcomes were measured only 4–6 weeks after the intervention. While improvements in knowledge, attitude, and behavior were statistically significant in the short

term, it remains unclear whether these gains persist over time without reinforcement. Prior research suggests that retention and behavioral adherence can diminish without ongoing engagement. Future work should include follow-up assessments at 6 months, 12 months, or longer to evaluate the durability of training effects and the need for refresher sessions or continuous awareness campaigns.

Addressing Demographic and Contextual Differences:

Although improvements were observed across employee groups (teachers, administrators, support staff), this study did not deeply explore how demographic factors such as age, prior IT experience, or technology attitudes may moderate training effectiveness. Likewise, contextual differences, such as urban versus rural districts or resource availability, were not systematically examined. Future research could stratify participants by these characteristics to identify potential disparities in learning outcomes and tailor training interventions to meet the needs of different subgroups. Additionally, cross-district studies could evaluate whether specific instructional methods (e.g., gamified modules, peer mentoring) are more effective for particular demographic or contextual profiles.

No One Left Behind: The fact that all subgroups of employees improved similarly is an encouraging sign for inclusivity of the training. It suggests that the content was accessible and relevant to both technical and non-technical staff, high-ranking and rank-and-file employees alike. Well-crafted training using plain language, real-life examples, and interactive elements can overcome typical barriers.

Behavior Change and Culture: One of the standout results was the increase in actual incident reporting (Figure 1), a strong indicator of behavior change. Employees not only reported more phishing attempts, but also demonstrated increased vigilance and responsibility. Over time, such cultural change can be self-reinforcing. Management support – in our case, the district’s encouragement of the training and deployment of the reporting tool – also signals that security is a priority, bolstering the culture.

Limitations: Despite positive outcomes, this study has some limitations that must be acknowledged. First, the research was conducted in a single school district, which may have unique characteristics (e.g., supportive leadership, specific demographic makeup) that aided program success. Second, the study relied on self-reported measures of behavior, which can introduce bias. We mitigated this by assuring anonymity and examining an objective measure (phishing reports). Third, the follow-up period was relatively short, as discussed above. Fourth, time constraints during the summer period may have limited some participants’ ability to fully engage with the material. Finally, our demographic analysis was limited and did not explore factors such as age, prior IT experience, or personal attitudes toward technology in depth.

V. CONCLUSION

This study provides robust evidence that a structured, interactive Information Security Awareness (ISA) training program can meaningfully enhance the knowledge, attitudes, and self-reported behaviors of K-12 school district employees. Following the intervention, staff demonstrated stronger understanding of security policies, more positive attitudes toward compliance, and increased engagement in secure practices, including a substantial rise in phishing incident reporting. These findings reinforce the validity of the Knowledge–Attitude–Behavior (KAB) model within an educational setting, showing that knowledge gains can translate into attitudinal shifts and tangible behavioral improvements.

While the study was conducted in a single California K-12 district, the underlying principles and training strategies are likely applicable to other educational environments and broader organizational contexts. The consistency of improvements across diverse employee roles – from teachers to administrators and support staff – suggests that well-designed, accessible, and interactive training can be effective for heterogeneous populations. Nevertheless, caution is warranted in generalizing results to organizations with different structural, cultural, or technological characteristics, as these factors may influence training outcomes. Future studies could explore variations across district sizes, resource levels, urban versus rural settings, and industries outside education to better understand context-dependent effectiveness.

An important consideration is that the improvements documented in this study were measured in the short term, 4–6 weeks after training. While statistically significant gains were observed, the durability of these effects over time remains uncertain. Knowledge and behavior retention may decay without reinforcement, highlighting the need for ongoing training, periodic refreshers, and continuous engagement strategies to sustain security-conscious behaviors.

The study also identifies opportunities for further exploration of demographic and contextual influences. Factors such as age, prior IT experience, technology attitudes, and departmental roles could moderate learning outcomes and behavior change. Tailoring training approaches to accommodate these differences – for example, using gamified modules, peer mentoring, or differentiated instruction – may optimize effectiveness across diverse staff populations. Additionally, integrating objective long-term behavioral metrics, such as simulated phishing exercises or automated incident tracking, would complement self-reported data and provide a more comprehensive view of sustained training

impact.

From a practical standpoint, this research underscores that investing in human-focused cybersecurity education is a critical component of organizational risk management. By equipping employees with both the knowledge and motivation to act securely, organizations can transform potential human vulnerabilities into a proactive first line of defense. The study further illustrates that combining interactive training with accessible tools – such as one-click phishing reporting mechanisms – strengthens the likelihood that awareness translates into action, fostering a culture of security vigilance.

In conclusion, the findings demonstrate that well-designed ISA programs can produce measurable improvements in security awareness, attitudes, and behaviors, even within complex and resource-constrained environments like K-12 education. When thoughtfully implemented, such programs not only enhance individual competence but also contribute to an organizational culture that values and prioritizes cybersecurity. By considering long-term retention, demographic and contextual variability, and ongoing reinforcement, future efforts can further optimize training strategies, extend their applicability across diverse sectors, and ensure that the human element strengthens, rather than undermines, overall security posture.

REFERENCES

- [1] I. Ajzen, N. Joyce, S. Sheikh, and N. G. Cote, “Knowledge and the Prediction of Behavior: The role of information accuracy in the theory of planned behavior,” *Basic and Applied Social Psychology*, vol. 33, pp. 101–117, 2011.
- [2] N. G. Arachchilage and S. Love, “Security awareness of computer users: A phishing threat avoidance perspective,” *Computers in Human Behavior*, pp. 304–312, 2014.
- [3] S. Aurangzeb, M. Aleem, M. Iqbal, and M. A. Islam, “Ransomware: A Survey and Trends,” *Journal of Information Assurance & Security*, vol. 6, no. 2, pp. 48–58, 2017.
- [4] J. B. Barlow, M. Warkentin, D. Ormond, and A. Dennis, “Don’t Even Think About It! The Effects of Antineutralization, Informational, and Normative Communication on Information Security Compliance,” *Journal of the Association for Information Systems*, vol. 19, no. 8, pp. 689–715, 2018.
- [5] B. Bulgurcu, H. Cavusoglu, and I. Benbasat, “Information security policy compliance: An empirical study of rationality-based beliefs and information security awareness,” *MIS Quarterly*, vol. 34, no. 3, pp. 523–548, 2010.
- [6] K. Dahbur, Z. Bashabsheh, and D. Bashabsheh, “Assessment of Security Awareness: A Qualitative and Quantitative Study,” *International Management Review*, vol. 13, no. 1, pp. 37–58, 2017.
- [7] M. Evans, L. Maglaras, Y. He, and H. Janicke, “Human behaviour as an aspect of cybersecurity assurance,” *Security & Communication Networks*, vol. 9, no. 17, pp. 4667–4679, 2016.
- [8] M. Fagan and M. Khan, “To follow or not to follow: A study of user motivations around cybersecurity advice,” *IEEE Internet Computing*, pp. 25–34, Sept./Oct. 2018.
- [9] W. R. Flores and M. Ekstedt, “Shaping intention to resist social engineering through transformational leadership, information security culture and awareness,” *Computers & Security*, vol. 59, pp. 26–44, 2016.
- [10] S. M. Furnell and J. D. Katsabas, “The challenges of understanding and using security: A survey of end-users,” *Computers & Security*, vol. 25, no. 1, pp. 27–35, 2005.
- [11] H. A. Kruger and W. D. Kearney, “A prototype for assessing information security,” *Computers & Security*, vol. 25, no. 4, pp. 289–296, 2006.
- [12] A. McCormac, T. Zwaans, K. Parsons, D. Calic, M. Butavicius, and M. Pattinson, “Individual differences and Information Security Awareness,” *Computers in Human Behavior*, vol. 69, pp. 151–156, 2016.
- [13] P. Meso, Y. Ding, and S. Xu, “The role of awareness and communications in information security management: A health care information systems perspective,” *International Journal of Management & Information Systems (Online)*, vol. 18, no. 2, pp. 47–67, 2013.
- [14] G. D. Moody, M. Siponen, and S. Pahnla, “Toward a unified model of information security policy compliance,” *MIS Quarterly*, vol. 42, no. 1, pp. 285–311, 2018.
- [15] A. Mylonas, A. Kastania, and D. Gritzalis, “Delegate the smartphone user? Security awareness in smartphone platforms,” *Computers & Security*, vol. 34, pp. 47–66, 2013.
- [16] K. Parsons, D. Calic, M. Pattinson, M. Butavicius, A. McCormac, and T. Zwaans, “The Human Aspects of Information Security Questionnaire (HAIS-Q): Two further validation studies,” *Computers & Security*, vol. 66, pp. 40–51, 2017.
- [17] K. Parsons, A. McCormac, M. Butavicius, M. Pattinson, and C. Jerram, “Determining employee awareness using the Human Aspects of Information Security Questionnaire (HAIS-Q),” *Computers & Security*, vol. 42, pp. 165–176, 2014.
- [18] P. S. Schrader and K. Lawless, “The Knowledge, Attitudes, & Behaviors Approach: How to Evaluate

Performance and Learning in Complex Environments,”
Performance Improvement, vol. 43, no. 9, pp. 8–15, 2004.

[19] J. Shropshire, M. Warkentin, and S. Sharma, “Personality, attitudes, and intentions: Predicting initial adoption of information security behavior,” *Computers & Security*, vol. 49, pp. 177–191, 2015.

[20] M. Warkentin and R. Willison, “Behavioral and policy issues in information systems security,” *European Journal of Information Systems*, vol. 18, no. 2, pp. 101–105, 2009.

[21] Q. Yi and N. Hohashi, “Comparison of perceptions of domestic elder abuse among healthcare workers based on the Knowledge-Attitude-Behavior (KAB) model,” *PLoS One*, vol. 13, 2018.